INDIA'S GLACIAL DISASTERS

We have had five major disasters showing the destructive power of a glacial flood

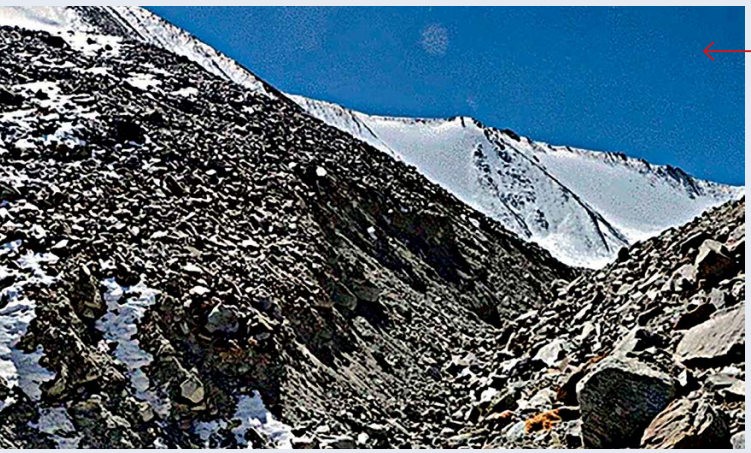


2013 KEDARNATH, *Uttarakhand*

CAUSE: Breach of the Chorabari Tal glacial lake, preceded by exceptionally heavy, sustained monsoon rainfall

AREAS AFFECTED: Mandakini and Alaknanda valleys, Rudraprayag district; Kedarnath town and pilgrim route worst hit

IMPACT: Nearly 6,000 people presumed dead; massive destruction of homes, roads, bridges across Uttarakhand



2014 GYA, *Ladakh*

CAUSE: Glacial Lake Outburst Flood (GLOF) at 5,400 m, caused by thawing of ice cores in the moraine

AREAS AFFECTED: Gya valley, Leh district and Kang Yatze massif near Taglang La; flood ran 16 km down the valley to the Leh–Manali highway

IMPACT: Agricultural terraces and fields were eroded; a concrete bridge, footbridge, irrigation canal intakes, two watermills and two houses were destroyed. The Leh–Manali highway was partly damaged

The warming trend is loading every part of this system. Across the Himalayas, temperatures have risen at roughly 0.15°C to 0.60°C a decade in the 21st century, faster at the greatest elevations. An analysis by Climate Trends, a research-based consulting and capacity-building initiative, found average Himalayan snow-surface temperatures on the Indian side increasing from -11.27°C in 2000-09 to -7.13°C in 2020-23. Black carbon from fossil-fuel use and biomass-burning in the Indo-Gangetic Plain darkens snow, increases heat absorption and accelerates melting, particularly in the central and eastern Himalayas.

What is visible from space is the retreating ice. The more insidious change occurs inside the mountain. As a glacier shrinks, it removes the support it once gave adjoining slopes, a process scientists call de-buttressing. At the same time, permafrost that filled fractures in the bedrock like cement begins to thaw. Meltwater enters those cracks, widens them through freezing and thawing, and can lubricate the surface between rock and ice. "Bedrock is glued together through cracks where frozen ice, or permafrost, is available," Haritashya explains.



"ISRO has given the govt detailed summary of Himalayan glacial lakes, with downstream risk assessment. But are these reports studied when planning urban projects?"

ANIL KULKARNI, *IISc, Bengaluru*